

FGFR1 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes FGFR1 gene fusions from the msk_impact_50k_2026 study, covering 23 fusion events. 7/23 fusions (30.4%) were in-frame. 11/23 fusions (47.8%) retained the PF07714 domain. Domain retention was tested with Fisher's exact test ($p=0.180728$, not statistically significant at $\alpha=0.05$). No literature benchmark is configured for FGFR1.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 23 analyzed fusion events, identifying 16 distinct fusion partner genes. The most frequent partner was NSD3, observed in 4 events.

Domain retention

PF07714 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 5/7 (71.4%) of in-frame fusions retained the domain, $p=0.180728$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.980198.

Domain disruption

Disruption of the configured disruption-required domain(s) was tested with Fisher's exact test comparing in-frame fusions against all others; 3/7 (42.9%) of in-frame fusions disrupted the domain, $p=0.627709$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.257426.

Cutpoint detection

Cutpoint detection scanned 22 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 471 aa (permutation-corrected $p=0.044955$, statistically significant at $\alpha=0.05$). This is -38 aa from the nearest configured domain boundary at 509 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=11$) and "not_retained" ($n=11$). For Frame_status, retained: 5/5 (100.0%, 95% CI 56.6%-100.0%); not_retained: 2/7 (28.6%, 95% CI 8.2%-64.1%). Welch's t-test comparing Tumor_variant_count between groups gave means of 78.9091 vs 30.8182, $p=0.347136$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score produced results for this run (16 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: FGFR1 has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
IGHMBP2	1	0.043478260869565216	0.0008686179185092667	0.05893480258118246	1.0	0.5033354555267505	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.2253811388094084	1
RAB11FIP1	1	0.043478260869565216	0.0008686179185092667	0.05893480258118246	0.9683604985618408	0.5033354555267505	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.22063521359368454	2
IFT140	1	0.043478260869565216	0.0008686179185092667	0.05893480258118246	0.8935762224352828	0.5033354555267505	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.20941757217470083	3
ADAM32	2	0.08695652173913043	0.0008686179185092667	0.05893480258118246	0.6735378715244487	0.5033354555267505	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1894552977989453	4
NSD3	4	0.17391304347826086	0.0008686179185092667	0.05893480258118246	0.45925215723873447	0.5033354555267505	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.18339939717782727	5
TNFRSF13B	1	0.043478260869565216	0.0008686179185092667	0.05893480258118246	0.6692233940556088	0.5033354555267505	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.17576464791774973	6
LETM2	1	0.043478260869565216	0.0008686179185092667	0.05893480258118246	0.6347075743048898	0.5033354555267505	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1705872749551419	7
TACC1	2	0.08695652173913043	0.0008686179185092667	0.05893480258118246	0.3974113135186961	0.5033354555267505	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1480363140980824	8
LPCAT3	1	0.043478260869565216	0.0008686179185092667	0.05893480258118246	0.4506232023010547	0.5033354555267505	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.14297461915456663	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
PLPBP	1	0.043478260869565216	0.0008686179185092667	0.05893480258118246	0.37008628954937683	0.5033354555267505	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.13089408224181495	10
PARP8	1	0.043478260869565216	0.0008686179185092667	0.05893480258118246	0.32981783317353786	0.5033354555267505	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.1248538137854391	11
CPM	1	0.043478260869565216	0.0008686179185092667	0.05893480258118246	0.30105465004793863	0.5033354555267505	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.12053933631659922	12
ACP3	3	0.13043478260869565	0.0008686179185092667	0.05893480258118246	0.0	0.5033354555267505	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.10146809533114756	13
ADGRE3	1	0.043478260869565216	0.0008686179185092667	0.05893480258118246	None	0.5033354555267505	['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']	0.08868369271695109	14
TAF4	1	0.043478260869565216	0.0008686179185092667	0.05893480258118246	0.08533077660594446	0.5033354555267505	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.08818075530030009	15
HOOK3	1	0.043478260869565216	0.0008686179185092667	0.05893480258118246	0.004793863854266611	0.5033354555267505	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.07610021838754842	16

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
64	2	9	0	11	None	0.47619047619047616	0.3222192947339193
153	4	7	1	10	5.714285714285714	0.3107769423558897	0.5075512105245131
238	4	7	2	9	2.5714285714285716	0.6351172047766476	0.19714612243907814
280	6	5	2	9	5.4	0.1826625386996904	0.7383505106889587
434	7	4	2	9	7.875	0.08049535603715172	1.0942291743602848
435	7	4	3	8	4.666666666666667	0.19838056680161942	0.7025008732311521
460	8	3	3	8	7.111111111111111	0.08610893750212636	1.064951769493942

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
471	9	2	3	8	12.0	0.02997312285237982	1.5232680062401485
566	9	2	4	7	7.875	0.08049535603715169	1.094229174360285
586	9	2	8	3	1.6875	1.0	-0.0
598	9	2	9	2	1.0	1.0	-0.0
690	9	2	10	1	0.45	1.0	-0.0
714	9	2	11	0	0.0	0.47619047619047616	0.3222192947339193
795	10	1	11	0	0.0	1.0	-0.0

domain_disruption tables

frame_domain_contingency_table

3	6
4	9

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf07679	Immunoglobulin I-set domain	22	13	1	8	0.05485232067510549

domain_retention tables

frame_domain_contingency_table

5	6
2	9

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein tyrosine and serine/threonine kinase	22	11	8	3	0.35838529701345584

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
11	16	23	0.6956521739130435

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ACP3	3
ADAM32	2
ADGRE3	1
CPM	1
HOOK3	1
IFT140	1
IGHMBP2	1
LETM2	1
LPCAT3	1
NSD3	4
PARP8	1
PLPBP	1
RAB11FIP1	1
TACC1	2
TAF4	1
TNFRSF13B	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0030267-T01-IM6-2	P-0030267-T01-IM6		CPM-FGFR1	CPM	out-of-frame	five_prime	38272053	16	714	True	disrupted	0.7436823104693141	True	Immunoglobulin-set domain		Protein tyrosine and serine/threonine kinase	CPM (NM_001005502) - FGFR1 (NM_001174067) rearrangement: t(8;12)(q11.23;q15)(chr8:g.38272053::chr12:g.69274149) Protein Fusion: out of frame {FGFR1:CPM}	NA

event_id	sampl e_id	pa tie nt _i d	fusion_ name	part ner _ge ne	fra me _st atu s	tar get _ro le	breakp oint_ge nomic	break point _exo n	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ dom ains	disrupt ed_do mains	source_annotation_text	source_site 2_effect_on _frame
EVT-P-000 4600-T01-I M5-4	P-0004 600-T0 1-IM5		FGFR1 -ACP3	ACP 3	in-fr ame	thr ee _pr im e	382860 54	5	153	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase; Immun oglobu lin I-set d omain			ACPP (NM_001134194) - FGFR1 (NM_001174067) rearrangement :t(3;8)(3q22 .1;8p11.33)(chr3:g.132081 259::chr8:g.38286054) Protein fusion: in frame (ACPP-FGFR1)	NA
EVT-P-000 4600-T02-I M5-5	P-0004 600-T0 2-IM5		FGFR1 -ACP3	ACP 3	in-fr ame	thr ee _pr im e	382860 54	5	153	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase; Immun oglobu lin I-set d omain			ACPP (NM_001134194) - FGFR1 (NM_001174067) t (3;8)(q22.1;p11.22)(chr3:g. 132081259::chr18:g.38286 054) Protein fusion: in frame (ACPP-FGFR1)	NA
EVT-P-002 3866-T01-I M6-6	P-0023 866-T0 1-IM6		FGFR1 -ACP3	ACP 3	in-fr ame	thr ee _pr im e	382875 77	4	64	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase; Immun oglobu lin I-set d omain			ACPP (NM_001134194) -FGFR1 (NM_001174067) rearrangement: t(3;8)(q22. 1;p11.23)(chr3:g.13207876 1::chr8:g.38287577) Protein Fusion: in frame {ACPP:FGFR1}	NA
EVT-P-002 1893-T01-I M6-7	P-0021 893-T0 1-IM6		FGFR1 -ADAM 32	ADA M32	in-fr ame	thr ee _pr im e	382836 16	8	280	True	retai ned	1.0	False	Protein tyrosin e and serine/ threoni ne kinase	Imm unogl obuli n I-set dom ain		ADAM32 (NM_145004) - FGFR1 (NM_001174067) rearrangement : c.58+3617 :ADAM32_c.838+24:FGFR 1inv Protein Fusion: in frame {ADAM32:FGFR1}	NA
EVT-P-003 1403-T01-I M6-8	P-0031 403-T0 1-IM6		FGFR1 -ADAM 32	ADA M32	unk now n	five _pr im e	382771 24	10	435	False	lost	0.0	False	Immun oglobu lin I-set d omain	Prote in tyr osine and s erine /thre onine kinas e		FGFR1 (NM_001174067) - ADAM32 (NM_145004) Rearrangement : c.1304:F GFR1_c.1233+2086:ADA M32 Protein Fusion: mid-exon {FGFR1:ADAM32}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0006645-T02-IM6-43	P-0006645-T02-IM6		FGFR1-HOOK3	HOOK3	unknown	five_prime	38271259	19	817	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin I-set domain			FGFR1 (NM_001174067) - HOOK3 (NM_032410) fusion: c.2449:FGFR1_c.268-255:HOOK3inv Protein Fusion: mid-exon {FGFR1:HOOK3}	NA
EVT-P-0058014-T02-IM7-44	P-0058014-T02-IM7		FGFR1-LETM2	LETM2	unknown	five_prime	38273543	14	598	False	disrupted	0.3249097472924188	True	Immunoglobulin I-set domain		Protein tyrosine and serine/threonine kinase	FGFR1 (NM_001174067) - LETM2 (NM_001199659) rearrangement: c.1792:FGFR1_c.1078-585:LETM2inv Protein Fusion: mid-exon {FGFR1:LETM2}	NA
EVT-P-0011515-T02-IM6-45	P-0011515-T02-IM6		FGFR1-PARP8	PARP8	out-of-frame	five_prime	38285268	6	238	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase	Immunoglobulin I-set domain	FGFR1 (NM_001174067) - PARP8 (NM_024615) rearrangement: t(5;8)(q11.1;p11.23)(chr5:g.50080979::chr8:g.38285268) Protein Fusion: out of frame {FGFR1:PARP8}	NA
EVT-P-0055303-T02-IM6-48	P-0055303-T02-IM6		FGFR1-TACC1	TACC1	unknown	five_prime	38274883	13	566	False	disrupted	0.20938628158844766	True	Immunoglobulin I-set domain		Protein tyrosine and serine/threonine kinase	FGFR1 (NM_001174067) - TACC1 (NM_001146216.2) fusion: c.1697:FGFR1_c.-425+18236:TACC1inv Protein Fusion: mid-exon {FGFR1:TACC1}	NA
EVT-P-0047086-T01-IM6-49	P-0047086-T01-IM6		FGFR1-TACC1	TACC1	in-frame	five_prime	38271378	18	795	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin I-set domain			FGFR1 (NM_001174067) - TACC1 (NM_006283) fusion: c.2386-56:FGFR1_c.1714-2100:TACC1inv Protein Fusion: in frame {FGFR1:TACC1}	NA

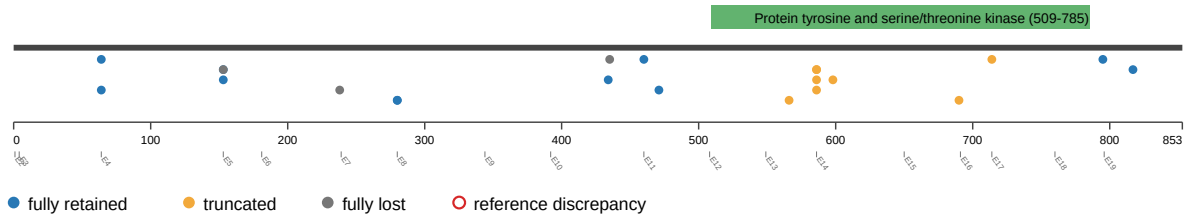
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0058899-T02-IM7-51	P-0058899-T02-IM7		FGFR1-TAF4	TAF4	in-frame	five_prime	38287091	4	153	True	lost	0.0	False		Protein tyrosine and serine/threonine kinase; Immunoglobulin I-set domain		FGFR1 (NM_001174067) - TAF4 (NM_003185) rearrangement: t(8;20)(p11.23;q13.33)(chr8:g.38287091::chr20:g.60602889) Protein Fusion: in frame {FGFR1:TAF4}	NA
EVT-P-0051005-T03-IM7-52	P-0051005-T03-IM7		IFT140-FGFR1	IFT140	unknown	three_prime	38277127	10	434	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Immunoglobulin I-set domain		IFT140 (NM_014714) - FGFR1 (NM_001174067) fusion: t(8;16)(p11.23;p13.3)(chr8:g.38277127::chr16:g.1576664) Protein Fusion: mid-exon {IFT140:FGFR1}	NA
EVT-P-0049789-T01-IM6-53	P-0049789-T01-IM6		IGHMBP2-FGFR1	IGHMBP2	unknown	three_prime	38275857	11	471	False	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Immunoglobulin I-set domain		IGHMBP2 (NM_002180) - FGFR1 (NM_001174067) fusion: t(8;11)(p11.23;q13.3)(chr8:g.38275857::chr11:g.68697608) Protein Fusion: mid-exon {IGHMBP2:FGFR1}	NA
EVT-P-0027047-T01-IM6-54	P-0027047-T01-IM6		LPCAT3-FGFR1	LPCAT3	unknown	three_prime	38282317	8	280	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Immunoglobulin I-set domain		LPCAT3 (NM_005768.5) - FGFR1 (NM_001174067) rearrangement: t(8;12)(p11.23;p13.31)(chr8:g.38282317::chr12:g.7127535) Transcript Fusion {LPCAT3:FGFR1}	NA
EVT-P-0025719-T01-IM6-55	P-0025719-T01-IM6		NSD3-FGFR1	NSD3	unknown	three_prime	38294436	4	64	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase; Immunoglobulin I-set domain			WHSC1L1 (NM_023034) - FGFR1 (NM_001174067) rearrangement: c.2555:WHSC1L1_c.191-6970:FGFR1 dup Protein Fusion: mid-exon {WHSC1L1:FGFR1}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-002 5390-T02-I M6-56	P-0025 390-T0 2-IM6		NSD3- FGFR1	NSD3	out-of-frame	five_prime	38274750	13	586	True	disrupted	0.2815884476534296	True	Immunoglobulin I-set domain		Protein tyrosine and serine/threonine kinase	FGFR1 (NM_001174067) - WHSC1L1 (NM_023034) rearrangement: c.1756+74:FGFR1_c.675+3313:WHSC1L1del Protein Fusion: out of frame {FGFR1:WHSC1L1}	NA
EVT-P-002 5390-T01-I M6-57	P-0025 390-T0 1-IM6		NSD3- FGFR1	NSD3	out-of-frame	five_prime	38274750	13	586	True	disrupted	0.2815884476534296	True	Immunoglobulin I-set domain		Protein tyrosine and serine/threonine kinase	FGFR1 (NM_001174067) - WHSC1L1 (NM_023034) rearrangement: c.1756+74:FGFR1_c.675+3313:WHSC1L1del Protein Fusion: out of frame {FGFR1:WHSC1L1}	NA
EVT-P-002 2996-T01-I M6-58	P-0022 996-T0 1-IM6		NSD3- FGFR1	NSD3	out-of-frame	three_prime	38274677	14	586	True	disrupted	0.7220216606498195	True		Immunoglobulin I-set domain	Protein tyrosine and serine/threonine kinase	WHSC1L1 (NM_023034) - FGFR1 (NM_001174067) rearrangement: c.1986+169:WHSC1L1_c.1756+147:FGFR1dup Protein Fusion: out of frame {WHSC1L1:FGFR1}	NA
EVT-P-002 8937-T01-I M6-59	P-0028 937-T0 1-IM6		PLPBP- FGFR1	PLPBP	unknown	five_prime	38272238	15	690	True	disrupted	0.6570397111913358	True	Immunoglobulin I-set domain		Protein tyrosine and serine/threonine kinase	FGFR1 (NM_001174067) rearrangement: c.2070+59:FGFR1_chr8:g.37619984del Antisense Fusion	NA
EVT-P-000 6646-T01-I M5-61	P-0006 646-T0 1-IM5		RAB11- FIP1-FGFR1	RAB11FIP1	unknown	three_prime	38275956	11	460	True	retained	1.0	False	Protein tyrosine and serine/threonine kinase	Immunoglobulin I-set domain		RAB11FIP1 (NM_001002814) - FGFR1 (NM_001174067) rearrangement : c.2027:RAB11FIP1_c.1378-65:FGFR1dup Protein fusion: mid-exon (RAB11FIP1-FGFR1)	NA
EVT-P-005 3401-T02-I M6-62	P-0053 401-T0 2-IM6		TNFRSF13B- FGFR1	TNFRSF13B	in-frame	three_prime	38273772	14	586	True	disrupted	0.7220216606498195	True		Immunoglobulin I-set domain	Protein tyrosine and serine/threonine kinase	TNFRSF13B (NM_012452) - FGFR1 (NM_001174067) rearrangement: t(8;17)(p11.23;p11.2)(chr8:g.38273772::chr17:g.16853401) Protein Fusion: in frame {TNFRSF13B:FGFR1}	NA

Figures

domain retention outliers

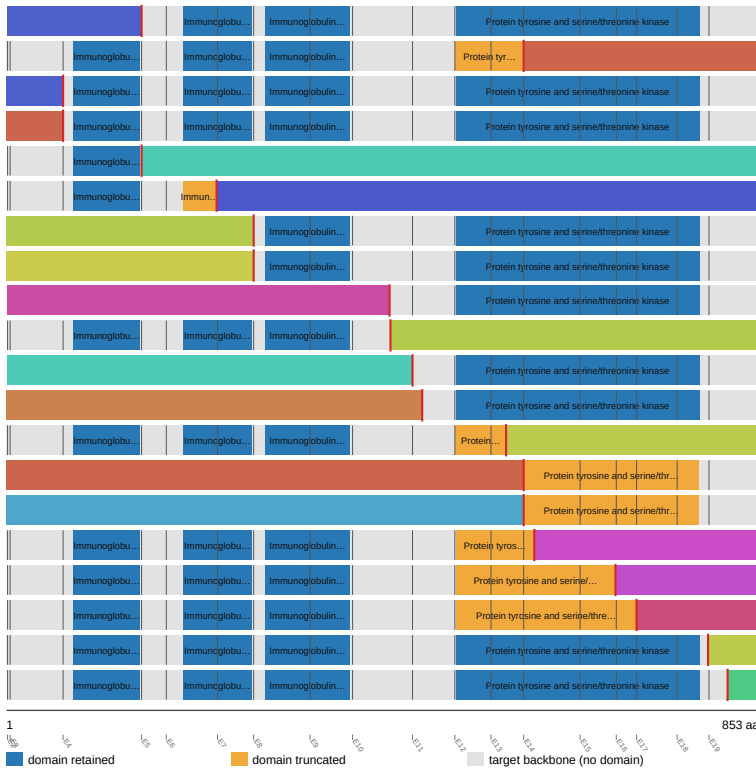
FGFR1 domain-retention track



fusion schematic

FGFR1 fusion-transcript schematic

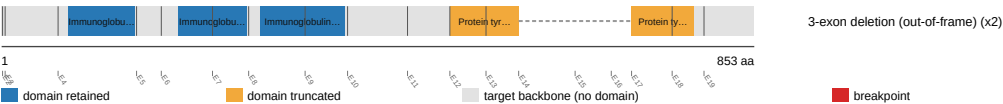
partner block → breakpoint → retained FGFR1 portion (domain-colored, exon ticks)



intragenic deletion schematic

FGFR1 intragenic-deletion schematic

retained N-terminal block → deleted span (plain connector) → resumed C-terminal block



reference comparison

Reference vs msk_impact_50k_2026

